Which City Has More Precipitation? Seattle or Miami?

Introduction

Seattle is widely notorious for its rainy, gloomy weather, but that notoriety does not necessarily mean that it receives more rainfall than other cities. For this project, I compared precipitation patterns in Seattle, Washington and Miami, Florida. I decided to choose Miami because South Florida experiences wet seasons and is affected by tropical weather systems such as hurricanes, making Miami an interesting candidate city to compare to Seattle.

The goal of my analysis was to determine whether it rains more in Seattle than Miami. Because "more rain" can refer to how often precipitation occurs and how much precipitation falls. Rainfall frequency and amount were both considered.

Methods

Daily precipitation data for Seattle and Miami from 2018 through 2022 were cleaned, organized and combined into a single dataset. The missing values found were managed before the two cities were compared.

My analysis examined average daily precipitation by month as well as the proportion of days with measurable precipitation. T and Z tests were used to determine whether differences between two cities were large enough to be considered meaningful rather than random variance data.

Results

The analysis suggested clear differences in rainfall patterns between the two cities. My hypothesis was Seattle has greater average daily precipitation during the winter. The results showed in fact Seattle had greater average precipitation during winter, particularly in January and December. Miami, on the other hand, had greater average precipitation during late spring and summer. Specifically from May through September.

The results showed that Seattle experiences more precipitation on more days during the majority of the year. This supports Seattle's reputation for frequent rainy weather. Interestingly, Miami's rainfall pattern is different: the data showed that rain occurred less frequently during some parts of the year, but rainfall during the wet season, the amount was often greater.

Statistical testing showed significant differences in mean daily precipitation during several months of the year, and precipitation frequency contrasted between cities during most months.

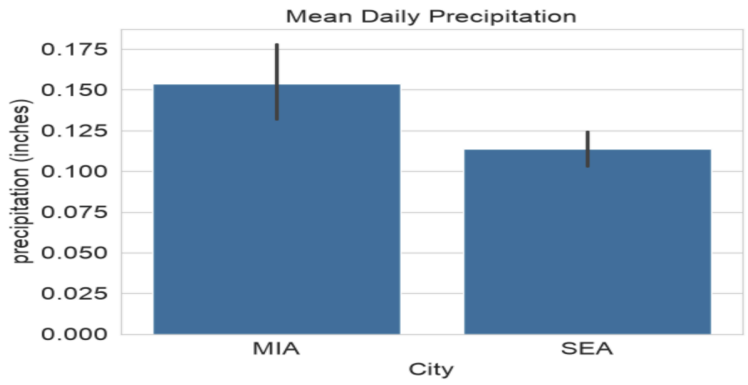


Figure 1. Mean daily precipitation by month in Seattle and Miami, 2018-2022.

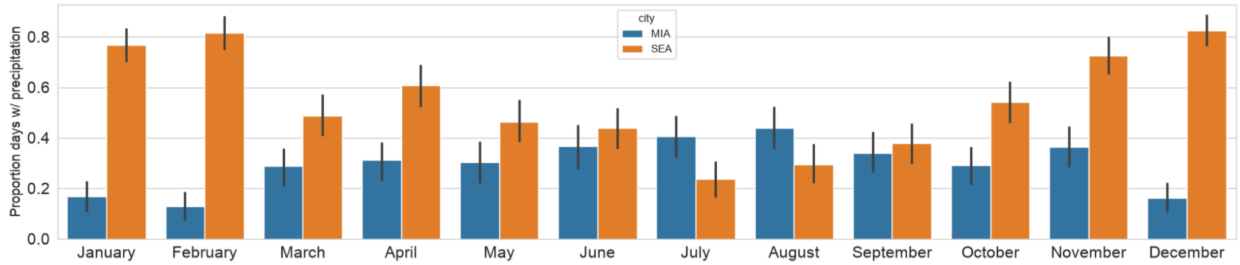


Figure 2. Proportion of Days with Precipitation by Month in Seattle and Miami, 2018-2022

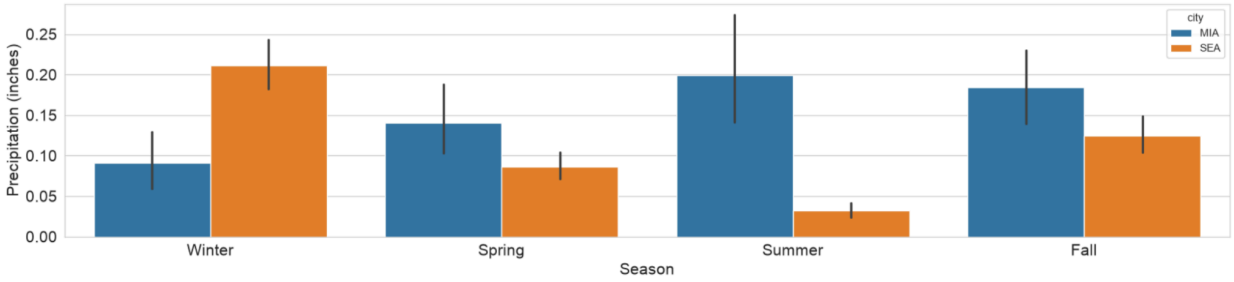


Figure 3. Mean Precipitation by Season in Seattle and Miami, 2018-2022

Conclusion

The results show that the conclusion that “Seattle gets more rain” than Miami is misleading. The difference is that Seattle tends to experience rain on a more frequent basis, especially during the cooler months, while Miami tends to receive larger amounts of rainfall during the wet seasons. Therefore, whether Seattle is considered to be more rainy, depends on how we measure rainfall. Seattle is rainier in terms of frequency, while Miami can be more rainy in terms of precipitation amount.